

## Discourse status as a cue to the acquisition of the Mandarin belief verb *yiwei*

Nonfactives, like Mandarin *juede* (1), are compatible with both true and false beliefs and are attested across languages. In contrast, contrafactuals, like Mandarin *yiwei* (2)[1], are more informative because they encode false beliefs, facilitating theory-of-mind reasoning[2-4], yet are rare crosslinguistically[5]. Belief verbs are generally difficult to acquire[6-8], and contrafactuals pose an additional challenge, as they may require learners to track that the embedded proposition *p* is false in context. Nevertheless, Mandarin-speaking children understand the contrafactivity of *yiwei p* by age four[4], so how do they acquire it?

We propose that discourses highlighting the falsity of *p* facilitate the acquisition of *yiwei*: When false beliefs are salient, caregivers may be more likely to comment on them[9], and *yiwei* is more likely to be used in these contexts than nonfactives[2]. More generally, context is often leveraged during word learning: violated expectations can help children acquire abstract terms like negation[9]; discourse relations can constrain referents of concrete words[10]. To infer belief verbs, adults rely on scenarios that make beliefs salient[11] alongside syntactic structures[6-8]. Yet, little is known about whether learners use the discourse status of *p* to distinguish different belief verbs.

We used the Human Simulation Paradigm with discourses containing *yiwei p* and *juede p* extracted from CHILDES[12]. Discourses differed in whether they provided information about the status of *p* (**Discourse-type**: constrained/unconstrained, Tab.1). We removed *yiwei/juede* from the target sentence, and participants either selected (2AFC;Exp.1) or reproduced (open-ended;Exp.2) the original verb. In each study, participants saw the target sentence either within its discourse or in isolation (**Context**: presence/absence). Responses matching the original verb were coded as correct (Fig.1). If discourse serves as a cue for *yiwei*, accuracy should be highest in the presence of a constrained discourse, since only this condition contains salient information that *p* is false.

For each experiment, we fit a logistic mixed-effects model predicting accuracy from **Verb**, **Discourse-type**, **Context**, and their interactions. In Exp.1, the main effect of **Context** ( $p<.001$ ) and its interaction with **Discourse-type** ( $p=.001$ ) were significant. Accuracy decreased in the absence of context, particularly in constrained conditions, suggesting that these discourse contexts contained information relevant to distinguishing the verbs.

Participants performed near ceiling in Exp.1, possibly due to the 2AFC. With the open-ended task in Exp.2, overall accuracy was substantially lower. Besides the significant effect of **Context** and its interaction with **Discourse-type**, Exp.2 revealed an interaction between **Verb** and **Context** ( $p<.001$ ), where *yiwei* had higher accuracy than *juede* in the presence of context. The significant three-way interaction ( $p=.018$ ) further indicated that the difference in accuracy between the verbs was larger in the constrained than in the unconstrained condition in the presence of context, but no such difference was observed in its absence.

These findings suggest that discourse information can serve as an effective cue for acquiring *yiwei*. Furthermore, the discourse status of *p* might help learners differentiate belief verbs. The systematic availability of such information in children's input remains an open question. Ongoing work investigates discourses in which *yiwei* and other belief verbs occur in child-directed speech.

(500 words)

- (1) Xiaoming juede Jiafei shi yizhi mao.  
 “Xiaoming thinks that Garfield is a cat.”  
 → Garfield **may or may not be** a cat.
- (2) Xiaoming yiwei Jiafei shi yizhi mao.  
 “Xiaoming falsely believes that Garfield is a cat.”  
 → Garfield **is not** a cat.

Table 1. English translation for example dialogues in each condition; the constraining discourse cues are bolded.

|              | Constrained ( $p$ for <i>juede</i> and $\neg p$ for <i>yiwei</i> )                                                                                                                                                                        | Unconstrained (no information to determine the truth value of $p$ )                                                                                                                                                                                                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>yiwei</i> | ADU: This was really well done.<br>ADU: Who made this?<br><b>CHI: (This was) made by Uncle.</b><br>ADU: Oh!<br>AUD: It was Uncle who made it?<br>ADU: I <u>yiwei</u> it was made by Grandma.<br>ADU: Such a beautiful car.<br>ADU: Oh no! | CHI: The kitten also wanted to play outside.<br>MOT: Look, this owner put up a missing cat poster.<br>CHI: Okay.<br>MOT: Let me read it to you. Lost cat notice. Okay?<br>MOT: You disappeared.<br>MOT: I <u>yiwei</u> it was just our usual hide-and-seek game.<br>MOT: You must be hiding in some corner, secretly laughing at me.<br>MOT: This is the fifth day. You’ve been hiding for so long. |
| <i>juede</i> | MOT: Is it a mosquito?<br>CHI: NO.<br>MOT: Then what is it?<br>CHI: A ladybug.<br><b>MOT: Oh, it is a ladybug.</b><br>MOT: You <u>juede</u> it is a ladybug.<br>MOT: The spider didn’t respond.<br>MOT: He’s busy spinning the web.       | MOT: Look, doesn’t it look like an egg?<br>MOT: How to put it together?<br>CHI: Put it together like this!<br>MOT: Put it together like this!<br>MOT: Right.<br>MOT: I also <u>juede</u> this puzzle is a little hard.<br>MOT: Okay.<br>MOT: Ah, I know.                                                                                                                                            |

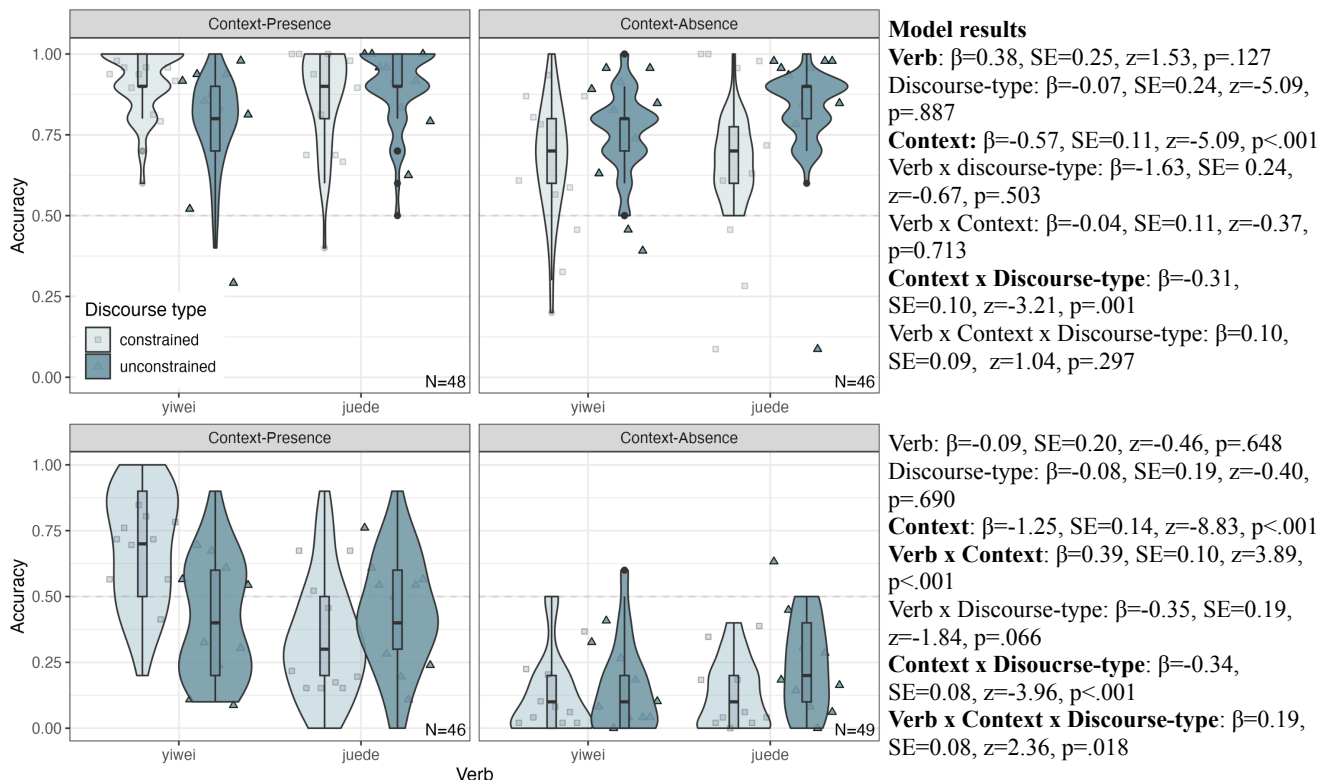


Figure 1. Mean accuracy of participants' guesses by verb and discourse type in Exp.1 (top) and Exp.2 (bottom).

**References:** [1] Glass (2025). [2] Lee et al. (1999). [3] Brandt, Li, & Chan (2023). [4] Zhang & Zhou (2022). [5] Holton (2017). [6] Gillette et al. (1999). [7] Snedeker & Gleitman (2004). [8] Gleitman et al. (2005). [9] Gomes, Huh, Yun, & Trueswell (2026). [10] Sullivan et al. (2019). [11] Papafragou, Cassidy, & Gleitman (2007). [12] Macwhinney (2000).